

01 GET /alarms

RULES ["compatibility aliases derive only from canonical fields", "source R03 requires alarm_type OOS, rule_code R03_CONSEC, and value null", "source SUMMARY requires rule_code SUMMARY_OOC", "source TRACE requires rule_code TRACE_OOS"]

```
{
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    "header": {},
    "path": {},
    "query": {
      "area": {
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        "required": false,
        "schema": {
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          "type": "string"
        },
        "chamber": {
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          "required": false,
          "schema": {
            "type": "string"
          },
          "date_from": {
            "nullable": true,
            "required": false,
            "schema": {
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              "type": "string"
            },
            "date_to": {
            "nullable": true,
            "required": false,
            "schema": {
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              "type": "string"
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            "required": false,
            "schema": {
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            },
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            "schema": {
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              "type": "boolean"
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            "required": false,
            "schema": {
              "type": "string"
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            "source": {
            "nullable": true,
            "required": true,
            "schema": {
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              "type": "string"
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                    "fields": {
                      "action_code": {
                        "nullable": true,
                        "required": false,
                        "schema": {
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                          "type": "string"
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                      "alarm_id": {
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                        "required": true,
                        "schema": {
                          "min_length": 1,
                          "type": "string"
                        },
                      "alarm_type": {
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                          "type": "string"
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                        "nullable": true,
                        "required": false,
                        "schema": {
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                          "type": "string"
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                        "nullable": true,
                        "required": false,
                        "schema": {
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                        "required": true,
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                          "type": "string"
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                        "schema": {
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                          "type": "string"
                        },
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                        "required": true,
                        "schema": {
                          "type": "string"
                        },
                      "mes_status": {
                        "nullable": true,
                        "required": false,
                        "schema": {
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                          "type": "string"
                        },
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                        "required": true,
                        "schema": {
                          "type": "boolean"
                        },
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                        "required": false,
                        "schema": {
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                          "type": "string"
                        },
                      "occurred_at": {
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                        "required": true,
                        "schema": {
                          "format": "date-time",
                          "type": "string"
                        },
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                        "nullable": false,
                        "required": true,
                        "schema": {
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                          "type": "string"
                        },
                      "parameter_id": {
                        "nullable": false,
                        "required": true,
                        "schema": {
                          "min_length": 1,
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                        },
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                        "required": false,
                        "schema": {
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                          "type": "string"
                        },
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                        "required": true,
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                        "required": true,
                        "schema": {
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                          "type": "string"
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                          "type": "integer"
                        },
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                        "nullable": false,
                        "required": true,
                        "schema": {
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                          "type": "string"
                        },
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                        "schema": {
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                          "type": "integer"
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                        "required": true,
                        "schema": {
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                          "type": "string"
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                        "schema": {
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                          "type": "integer"
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                        },
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                        "required": false,
                        "schema": {
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                        "required": true,
                        "schema": {
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                          "type": "string"
                        },
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                        "required": true,
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                          "min_length": 1,
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                              "nullable": false,
                              "required": false,
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                                            },
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                              }
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                          }
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                    }
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                }
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        }
      }
    }
  }
}
```

02 GET /trace

RULES []

```
{
  "request": {
    "body": null,
    "header": {},
    "path": {},
    "query": {
      "chamber": {
        "nullable": false,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        },
      "lot": {
        "nullable": false,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        },
      "parameter": {
        "nullable": false,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        },
      "wafer": {
        "nullable": false,
        "required": true,
        "schema": {
          "min_length": 1,
          "type": "string"
        }
      }
    },
    "responses": {
      "200": {
        "schema": {
          "items": {
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            "fields": {
              "measured_at": {
                "nullable": false,
                "required": true,
                "schema": {
                  "format": "date-time",
                  "type": "string"
                },
              "recipe_step_no": {
                "nullable": false,
                "required": true,
                "schema": {
                  "minimum": 1.0,
                  "type": "integer"
                },
              "seq_no": {
                "nullable": false,
                "required": true,
                "schema": {
                  "minimum": 0.0,
                  "type": "integer"
                },
              "value": {
                "nullable": false,
                "required": true,
                "schema": {
                  "type": "number"
                },
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                      "required": false,
                      "schema": {
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                          "fields": {
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                                    {
                                      "type": "integer"
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                                  ],
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                                    },
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                                  "type": "array",
                                  "shape": "object"
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                              }
                            }
                          }
                        }
                      }
                    }
                  }
                }
              }
            }
          }
        }
      }
    }
  }
}
```

03 GET /parameters

RULES []

```
{
  "request": {
    "body": null,
    "header": {},
    "path": {},
    "query": {},
    "responses": {
      "200": {
        "schema": {
          "item
```

```
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```

04 POST /documents/search

RULES []

```
{request: {body: {"additional_properties": false, "fields": {"doc_type": {"nullable": true, "required": false, "schema": {"enum": ["MANUAL", "SPEC", "TROUBLESHOOT"], "type": "string"}}, "model_code": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "query": {"nullable": false, "required": true, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, "top_k": {"nullable": false, "required": false, "schema": {"default": 4, "maximum": 10.0, "minimum": 1.0, "type": "integer"}}, "type": {"type": "object"}, "header": {}, "path": {}, "query": {}, "responses": {"200": {"schema": {"items": {"additional_properties": false, "fields": {"chunk_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "content": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "doc_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "document_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "model_code": {"nullable": true, "required": false, "schema": {"min_length": 1, "type": "string"}}, "score": {"nullable": false, "required": true, "schema": {"maximum": 1.0, "minimum": -1.0, "type": "number"}}, "section": {"nullable": true, "required": false, "schema": {"type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"type": "object"}, "array": true, "shape": "array", "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "type": "array"}}, "type": "object"}, "shape": "object"}}
```

05 GET /relations/chambers/{chamber_id}

RULES [{"node_count equals len(nodes)", "relationship_count equals len(relationships)"]

```
{request: {body: null, "header": {}, "path": {"chamber_id": {"nullable": false, "required": true, "schema": {"type": "string"}}, "query": {"include_production_context": {"nullable": false, "required": false, "schema": {"default": false, "type": "boolean"}}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {"graph_revision": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "nodes": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"business_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "display_name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "label": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "properties": {"nullable": false, "required": true, "schema": {"additional_properties": true, "fields": {"type": "object"}}, "type": "object"}, "type": "array"}}, "production_context": {"nullable": true, "required": false, "schema": {"additional_properties": false, "fields": {"returned_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "truncated": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "type": "object"}, "relationships": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "target": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}}, "root_node_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "shape": "object", "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "type": "array"}}, "type": "object"}, "shape": "object"}}
```

06 GET /agent/runs

RULES []

```
{request: {body: null, "header": {}, "path": {}, "query": {"date_from": {"nullable": true, "required": false, "schema": {"format": "date", "type": "string"}}, "date_to": {"nullable": true, "required": false, "schema": {"format": "date", "type": "string"}}, "predicted_fault_code": {"nullable": true, "required": false, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "status": {"nullable": true, "required": false, "schema": {"enum": ["COMPLETED", "FAILED", "RUNNING", "WAITING_APPROVAL"], "type": "string"}}, "responses": {"200": {"schema": {"items": {"additional_properties": false, "fields": {"action_id": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "alarm_id"
```

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  "nullable": false, "required": true, "schema": {"enum": ["R03", "SUMMARY", "TRACE"], "type": "string"}},
"approval_id": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber": {
  "nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber_id": {
  "nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "confidence": {"nu
  llable": true, "required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}, "created_
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  ies": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields"
    : {"channel": {"nullable": false, "required": true, "schema": {"enum": ["EMAIL", "MES"], "type": "string"
      }}, "status": {"nullable": false, "required": true, "schema": {"enum": ["BLOCKED", "CANCELED", "FAILED",
        "SENDING", "SENT", "UNKNOWN", "WAITING"], "type": "string"}}, "type": "object", "type": "array"}}, "fau
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              lse, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "llm_model": {"nullable": false, "
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                        "FAILED", "RUNNING", "WAITING_APPROVAL"], "type": "string"}}, "tools": {"nullable": false, "required": tr
                          ue, "schema": {"items": {"additional_properties": false, "fields": {"n": {"nullable": false, "required":
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                              rue, "schema": {"min_length": 1, "type": "string"}}, "s": {"nullable": false, "required": true, "schema":
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                                      y"}}, "type": "object", "type": "array", "shape": "array", "422": {"schema": {"additional_properties"
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                                          erties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "unio
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                                                {"type": "string"}}, "type": "object", "type": "array"}}, "type": "object", "shape": "object"}},
                                                07 POST /agent/ask
```

```
RULES [{"fault_code and ground_truth_fault_code are forbidden", "metrology.alarm_result is forbidden"}]
{"request": {"body": {"additional_properties": false, "fields": {"agent_run_id": {"nullable": true, "requ
  ired": false, "schema": {"min_length": 1, "type": "string"}}, "question": {"nullable": false, "required":
    true, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, "type": "object", "header":
      {}, "path": {}, "query": {}, "responses": {"200": {"schema": {"additional_properties": false, "fields":
        {"answer": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "confide
          nce": {"nullable": true, "required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}},
            "evidence": {"nullable": true, "required": true, "schema": {"additional_properties": false, "fields": {"
              chunk_id": {"nullable": false, "schema": {"min_length": 1, "type": "string"}}, "doc_id"
                : {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "document_id": {"
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                            le": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false
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                                            : 1, "type": "string"}}, "document_id": {"nullable": false, "required": true, "schema": {"min_length": 1,
                                              "type": "string"}}, "excerpt": {"nullable": false, "required": true, "schema": {"min_length": 1, "type":
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                                                                  lable": false, "required": true, "schema": {"type": "string"}}, "type": "object"}, "METROLOGY": {"additi
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```

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```

08 GET /approvals

RULES ["APPROVED or REJECTED requires decided_by and decided_at", "PENDING requires all decision fields not null", "only EQP_HOLD actions have approvals"]

```
{
  "request": {
    "body": null,
    "header": {},
    "path": {},
    "query": {},
    "responses": {
      "200": {
        "schema": {
          "item s": {
            "additional_properties": false,
            "fields": {
              "action_code": {
                "nullable": false,
                "required": true,
                "schema": {
                  "enum": [
                    "EQP_HOLD",
                    "MONITORING",
                    "WARNING"
                  ],
                  "type": "string"
                },
              "action_id": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
              "agent_run_id": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
              "approval_id": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
              "approved_at": {
                "nullable": true,
                "required": true,
                "schema": {
                  "format": "date-time",
                  "type": "string"
                },
              "approved_by": {
                "nullable": true,
                "required": true,
                "schema": {
                  "type": "string"
                },
              "chamber": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
              "chamber_id": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
              "created_at": {
                "nullable": false,
                "required": true,
                "schema": {
                  "format": "date-time",
                  "type": "string"
                },
              "decided_at": {
                "nullable": true,
                "required": true,
                "schema": {
                  "format": "date-time",
                  "type": "string"
                },
              "decided_by": {
                "nullable": true,
                "required": true,
                "schema": {
                  "type": "string"
                },
              "decision_comment": {
                "nullable": true,
                "required": true,
                "schema": {
                  "type": "string"
                },
              "equipment": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
              "equipment_id": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
              "fault_code": {
                "nullable": false,
                "required": true,
                "schema": {
                  "enum": [
                    "FOC",
                    "MFD",
                    "OTH",
                    "RFM",
                    "TMD"
                  ],
                  "type": "string"
                },
              "lot": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
              "lot_id": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
              "predicted_fault_code": {
                "nullable": false,
                "required": true,
                "schema": {
                  "enum": [
                    "FOC",
                    "MFD",
                    "OTH",
                    "RFM",
                    "TMD"
                  ],
                  "type": "string"
                },
              "reason": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
              "status": {
                "nullable": false,
                "required": true,
                "schema": {
                  "enum": [
                    "APPROVED",
                    "PENDING",
                    "REJECTED"
                  ],
                  "type": "string"
                },
              "type": "object",
              "type": "array",
              "shape": "array"
            }
          }
        }
      }
    }
  }
}
```

09 POST /approvals/{approval_id}/decision

RULES ["APPROVED or REJECTED requires decided_by and decided_at", "PENDING requires all decision fields not null", "only EQP_HOLD actions have approvals"]

```
{
  "request": {
    "body": {
      "additional_properties": false,
      "fields": {
        "decided_by": {
          "nullable": false,
          "required": true,
          "schema": {
            "max_length": 40,
            "min_length": 1,
            "type": "string"
          }
        },
        "decision": {
          "nullable": false,
          "required": true,
          "schema": {
            "enum": ["APPROVED", "REJECTED"],
            "type": "string"
          }
        },
        "decision_comment": {
          "nullable": true,
          "required": false,
          "schema": {
            "max_length": 1000,
            "type": "string"
          }
        },
        "object": {
          "header": {},
          "path": {
            "approval_id": {
              "nullable": false,
              "required": true,
              "schema": {
                "type": "string"
              }
            },
            "query": {},
            "responses": {
              "200": {
                "schema": {
                  "additional_properties": false,
                  "fields": {
                    "action_code": {
                      "nullable": false,
                      "required": true,
                      "schema": {
                        "enum": ["EQP_HOLD", "MONITORING", "WARNING"],
                        "type": "string"
                      }
                    },
                    "action_id": {
                      "nullable": false,
                      "required": true,
                      "schema": {
                        "min_length": 1,
                        "type": "string"
                      }
                    },
                    "agent_run_id": {
                      "nullable": false,
                      "required": true,
                      "schema": {
                        "min_length": 1,
                        "type": "string"
                      }
                    },
                    "approval_id": {
                      "nullable": false,
                      "required": true,
                      "schema": {
                        "min_length": 1,
                        "type": "string"
                      }
                    },
                    "approved_at": {
                      "nullable": true,
                      "required": true,
                      "schema": {
                        "format": "date-time",
                        "type": "string"
                      }
                    }
                  }
                }
              }
            }
          }
        }
      }
    }
  }
}
```

```
ring"}}, "approved_by": {"nullable": true, "required": true, "schema": {"type": "string"}}, "chamber": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "created_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "decided_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "decided_by": {"nullable": true, "required": true, "schema": {"type": "string"}}, "decision_comment": {"nullable": true, "required": true, "schema": {"type": "string"}}, "equipment": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "fault_code": {"nullable": false, "required": true, "schema": {"enum": ["FOC", "MFD", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "lot": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "predicted_fault_code": {"nullable": false, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "reason": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["APPROVED", "PENDING", "REJECTED"], "type": "string"}}, "type": "object"}, "shape": "object"}, "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "array"}}, "type": "object"}, "shape": "object"}}
```

10 GET /audit-logs

RULES []

```
{"request": {"body": null, "header": {}, "path": {}, "query": {"actor_type": {"nullable": true, "required": false, "schema": {"type": "string"}}, "date_from": {"nullable": true, "required": false, "schema": {"format": "date", "type": "string"}}, "date_to": {"nullable": true, "required": false, "schema": {"format": "date", "type": "string"}}, "entity_id": {"nullable": true, "required": false, "schema": {"type": "string"}}, "entity_type": {"nullable": true, "required": false, "schema": {"type": "string"}}, "event_type": {"nullable": true, "required": false, "schema": {"type": "string"}}}, "responses": {"200": {"schema": {"items": {"additional_properties": true, "fields": {"actor": {"nullable": false, "required": true, "schema": {"type": "string"}}, "actor_id": {"nullable": true, "required": true, "schema": {"type": "string"}}, "actor_type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "after": {"nullable": true, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "audit_id": {"nullable": false, "required": true, "schema": {"type": "integer"}}, "before": {"nullable": true, "required": true, "schema": {"additional_properties": true, "fields": {}, "type": "object"}}, "detail": {"nullable": true, "required": true, "schema": {"type": "string"}}, "entity": {"nullable": false, "required": true, "schema": {"type": "string"}}, "entity_id": {"nullable": false, "required": true, "schema": {"type": "string"}}, "entity_type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "event": {"nullable": false, "required": true, "schema": {"type": "string"}}, "event_type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "occurred_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "type": "object"}, "type": "array"}, "shape": "array"}, "422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}}, "type": "array"}}, "msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": "array"}}, "type": "object"}, "shape": "object"}}
```

11 GET /dataset/bounds

RULES []

deferred inventory

12 GET /dashboard/summary

RULES []

```
{"request": {"body": null, "header": {}, "path": {}, "query": {"area": {"nullable": true, "required": false, "schema": {"enum": ["ALL", "Etch", "Photo"], "type": "string"}}, "chamber_id": {"nullable": true, "required": false, "schema": {"type": "string"}}, "date": {"nullable": true, "required": false, "schema": {"format": "date", "type": "string"}}, "equipment_id": {"nullable": true, "required": false, "schema": {"type": "string"}}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {"alarm_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "area": {"nullable": true, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "daily_trend": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"date": {"nullable": false, "required": true, "schema": {"format": "date", "type": "string"}}, "has_r03_consec": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "ooc_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "oos_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "type": "object"}, "type": "array"}, "date_range": {"nullable": false, "required": true, "schema": {"items": {"format": "date", "type": "string"}, "type": "array"}}, "equipment_counts": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"alarm_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "area_id": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "chambers": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"alarm_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}}
```

```
{}}, "hierarchy": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"area_id": {"nullable": false, "required": true, "schema": {"enum": ["Etch", "Photo"], "type": "string"}}, "chambers": {"nullable": false, "required": true, "schema": {"items": {"min_length": 1, "type": "string"}, "type": "array"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}}, "ooc_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "oos_count": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "pending_approvals": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"action_code": {"nullable": false, "required": true, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "action_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approval_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "equipment_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "incident": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}}, "recent_alarms": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"action_code": {"nullable": true, "required": false, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "action_id": {"nullable": true, "required": false, 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13 GET /alarms/{source}/{alarm_id}

RULES []

```
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```

```
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14 GET /alarms/paged

RULES []

```
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```

15 GET /traces/catalog

RULES []

```
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```

```
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```

16 POST /traces/search

RULES []

```
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17 GET /relations/equipment/{equipment_id}

RULES []

deferred inventory

18 GET /documents/{document_id}

RULES []

```
{ "request": { "body": null, "header": {}, "path": { "document_id": { "nullable": false, "required": true, "schema": {"type": "string"} }, "query": {} }, "responses": { "200": { "schema": { "additional_properties": false, "fields": { "chunks": { "nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"chunk_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "array"} }, "type": "object"}, "shape": "object"} }
```



```
{
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19 POST /agent/runs
RULES []
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20 GET /agent/runs/{run_id}
RULES [delivery_policy_lub204lub77dubc740 ACTION-POLICY-V1 lub638lub658lub774luba70 MOCK-NOTIFY-V1lub740lub2b9lubc778lubc18lub824lub00b7lub655lubc778luae30lub85dluc5c6lub774lubc774luba54lub77clubcfubc790lub3d9Kafka/MESMocklubc744lubc5f0lub3d9lubd55club2e4;luc2b9lubc778lubc0clubd0dcclub294nulllubc774lub2e4]
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22 POST /agent/runs{run_id}/retry
RULES []
deferred inventory
23 GET /agent/runs/paged
RULES []
deferred inventory
24 GET /approvals/paged
RULES []
deferred inventory
25 GET /actions
RULES [{"delivery_policy \ub204\ub77d\uc740 ACTION-POLICY-V1 \ud638\ud658\uc774\uba70 MOCK-NOTIFY-V1\uc740
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b3d9 Kafka/MES Mock\uc744 \uc5f0\ub3d9\ud55c\ub2e4; \uc2b9\uc778 \uc0c1\ud0dc\ub294 null\uc774\ub2e4"}
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```

[illegible]

```
}}}
27 POST /actions/{action_id}/deliveries/{channel}/retry
RULES []
deferred inventory
28 POST /internal/actions/{action_id}/delivery
RULES [{"FAILED requires error_code non-null and permits provider_message_id null", "SENT requires provide
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29 POST /analytics/query
RULES []
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RULES ¶

31 POST /analytics/validate

32 GET /analytics/history

33 GET /analytics/evaluations

```

RULES []
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    "path": {},
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        "required": false,
        "schema": {
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          "type": "boolean"
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        "required": false,
        "schema": {
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          "minimum": 1,
          "type": "integer"
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      "size": {
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        "required": false,
        "schema": {
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          "maximum": 100,
          "minimum": 1,
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    },
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34 GET /audit-logs/paged

RULES []

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[illegible]

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AgentAskResponse

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AgentEvaluationResponse

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AgentRunAccepted

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AgentRunAskEvidence

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AgentRunDetailResponse

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AgentRunItem

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ApprovalItem

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{ "additional_properties": false, "fields": { "action_code": { "nullable": false, "required": true, "schema": { "enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string" } }, "action_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "agent_run_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "approval_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "approved_at": { "nullable": true, "required": true, "schema": { "format": "date-time", "type": "string" } }, "approved_by": { "nullable": true, "required": true, "schema": { "min_length": 1, "type": "string" } }, "chamber": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "chamber_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "created_at": { "nullable": false, "required": true, "schema": { "format": "date-time", "type": "string" } }, "decided_at": { "nullable": true, "required": true, "schema": { "format": "date-time", "type": "string" } }, "decided_by": { "nullable": true, "required": true, "schema": { "min_length": 1, "type": "string" } }, "decision_comment": { "nullable": true, "required": true, "schema": { "max_length": 1000, "min_length": 1, "type": "string" } }, "equipment": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "equipment_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "fault_code": { "nullable": false, "required": true, "schema": { "enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string" } }, "lot": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "lot_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "predicted_fault_code": { "nullable": false, "required": true, "schema": { "enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string" } }, "reason": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "status": { "nullable": false, "required": true, "schema": { "enum": ["APPROVED", "PENDING", "REJECTED"], "type": "string" } } }, "rules": ["PENDING requires all decision fields null", "APPROVED or REJECTED requires decided_by and decided_at", "only EQP_HOLD actions have approvals"], "type": "object" }
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AskDocumentEvidenceAlias

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AskToolItem

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{ "additional_properties": false, "fields": { "name": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "result": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "result_summary": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "status": { "nullable": false, "required": true, "schema": { "enum": ["ERROR", "SUCCESS", "TIMEOUT"], "type": "string" } }, "tool_name": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } } }, "type": "object" }
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AuditLogItem

```
{ "additional_properties": false, "fields": { "actor": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "actor_id": { "nullable": true, "required": true, "schema": { "type": "string" } }, "actor_type": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "after": { "nullable": true, "required": true, "schema": { "additional_properties": true, "fields": { "type": "object" } }, "at": { "nullable": false, "required": true, "schema": { "format": "date-time", "type": "string" } }, "audit_id": { "nullable": false, "required": true, "schema": { "minimum": 1, "type": "integer" } }, "before": { "nullable": true, "required": true, "schema": { "additional_properties": true, "fields": { "type": "object" } }, "detail": { "nullable": true, "required": true, "schema": { "type": "string" } }, "entity": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "entity_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "entity_type": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "event": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "event_type": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "occurred_at": { "nullable": false, "required": true, "schema": { "format": "date-time", "type": "string" } } }, "type": "object" }
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AuditLogPageResponse

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{ "additional_properties": true, "fields": { "event_type_counts": { "nullable": false, "required": true, "schema": { "additional_properties": {"type": "integer"}, "fields": {}, "type": "object" } }, "event_types": { "nullable": false, "required": true, "schema": { "items": {"type": "string"}, "type": "array" } }, "items": { "nullable": false, "required": true, "schema": { "items": { "additional_properties": true, "fields": { "actor": { "nullable": false, "required": true, "schema": { "type": "string" } }, "actor_id": { "nullable": true, "required": true, "schema": { "type": "string" } }, "actor_type": { "nullable": false, "required": true, "schema": { "type": "string" } }, "after": { "nullable": true, "required": true, "schema": { "additional_properties": true, "fields": {}, "type": "object" } }, "at": { "nullable": false, "required": true, "schema": { "format": "date-time", "type": "string" } }, "audit_id": { "nullable": false, "required": true, "schema": { "type": "integer" } }, "before": { "nullable": true, "required": true, "schema": { "additional_properties": true, "fi
```

```
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AutoToolCallItem

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ChamberGraphResponse

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ChamberRelationResponse

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{"additional_properties": false, "fields": {"graph_revision": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "nodes": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"business_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "display_name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "label": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "properties": {"nullable": false, "required": true, "schema": {"additional_properties": true, "fields": {"type": "object"}}, "type": "object"}, "type": "array"}, "relationships": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "target": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}, "type": "array"}, "root_node_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}
```

ChartType

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{"enum": ["bar", "histogram", "line", "table"], "type": "string"}
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ChatToolCallItem

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{"additional_properties": false, "fields": {"name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "result": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "result_summary": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["ERROR", "SUCCESS", "TIMEOUT"], "type": "string"}}, "tool_name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}
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ComparisonMatrix

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{"additional_properties": false, "fields": {"downstream": {"nullable": false, "required": false, "schema": {"default": "NOT_AVAILABLE", "enum": ["CHECKED", "NOT_AVAILABLE", "NOT_CHECKED"], "type": "string"}}, "history": {"nullable": false, "required": false, "schema": {"default": "NOT_AVAILABLE", "enum": ["CHECKED", "NOT_AVAILABLE", "NOT_CHECKED"], "type": "string"}}, "metrology": {"nullable": false, "required": false, "schema": {"default": "NOT_AVAILABLE", "enum": ["CHECKED", "NOT_AVAILABLE", "NOT_CHECKED"], "type": "string"}}, "type": "object"}
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e, "schema": {"default": "NOT_AVAILABLE", "enum": ["CHECKED", "NOT_AVAILABLE", "NOT_CHECKED"], "type": "string"}}, "sibling": {"nullable": false, "required": false, "schema": {"default": "NOT_AVAILABLE", "enum": ["CHECKED", "NOT_AVAILABLE", "NOT_CHECKED"], "type": "string"}}, "upstream": {"nullable": false, "required": false, "schema": {"default": "NOT_AVAILABLE", "enum": ["CHECKED", "NOT_AVAILABLE", "NOT_CHECKED"], "type": "string"}}, "type": "object"}
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e, "schema": {"pattern": "^A-Z0-9_+\$\$", "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["AVAILABLE", "EMPTY"], "type": "string"}}, "verification_steps": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}}, "type": "object"}

DocumentAskEvidence

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DocumentChunkItem

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DocumentDetailResponse

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DocumentEvidence

{"additional_properties": false, "fields": {"chunk_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "document_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "excerpt": {"nullable": false, "required": true, "schema": {"type": "string"}}, "section": {"nullable": true, "required": true, "schema": {"type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"enum": ["DOCUMENT"], "type": "string"}}, "type": "object"}

DocumentHit

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DocumentSearchRequest

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DocumentType

{"enum": ["MANUAL", "SPEC", "TROUBLESHOOT"], "type": "string"}

ErrorCode

{"enum": ["APPROVAL_ALREADY_DECIDED", "DEPENDENCY_NOT_READY", "IDEMPOTENCY_CONFLICT", "INCIDENT_ALREADY_PROCESSED", "INCIDENT_ALREADY_RUNNING", "INTERNAL_ERROR", "LEGACY_APPROVAL_NOT_LINKED", "LLM_NOT_READY", "MODEL_NOT_READY", "POLICY_REJECTED", "RESOURCE_NOT_FOUND", "UNAUTHORIZED", "VALIDATION_ERROR"], "type": "string"}

ErrorResponse

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EvaluationListResponse

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EvidenceAssessmentBlock
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EvidenceItem
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FaultHypothesis
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GraphAskEvidence
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```

GraphContext

```
{
  "additional_properties": false,
  "fields": {
    "adjacent_process_step_ids": {
      "nullable": false,
      "required": true,
      "schema": {
        "items": {
          "type": "string",
          "type": "array"
        },
        "area": {
          "nullable": false,
          "required": true,
          "schema": {
            "enum": ["Etch", "Photo"],
            "type": "string"
          },
          "chamber_id": {
            "nullable": false,
            "required": true,
            "schema": {
              "min_length": 1,
              "type": "string"
            },
            "equipment_id": {
              "nullable": false,
              "required": true,
              "schema": {
                "min_length": 1,
                "type": "string"
              },
              "model_code": {
                "nullable": false,
                "required": true,
                "schema": {
                  "min_length": 1,
                  "type": "string"
                },
                "parameter_ids": {
                  "nullable": false,
                  "required": true,
                  "schema": {
                    "items": {
                      "type": "string",
                      "type": "array"
                    },
                    "process_step_id": {
                      "nullable": false,
                      "required": true,
                      "schema": {
                        "min_length": 1,
                        "type": "string"
                      },
                      "relation_ids": {
                        "nullable": false,
                        "required": true,
                        "schema": {
                          "items": {
                            "type": "string",
                            "type": "array"
                          },
                          "type": "object"
                        }
                      }
                    }
                  }
                }
              }
            }
          }
        }
      }
    }
  }
}
```

GraphEvidence

```
{
  "additional_properties": false,
  "fields": {
    "excerpt": {
      "nullable": false,
      "required": true,
      "schema": {
        "type": "string"
      },
      "graph_revision": {
        "nullable": false,
        "required": true,
        "schema": {
          "pattern": "^0-9a-f{64}$",
          "type": "string"
        },
        "relation_id": {
          "nullable": false,
          "required": true,
          "schema": {
            "pattern": "^REL-[0-9a-f]{20}$",
            "type": "string"
          },
          "source_id": {
            "nullable": false,
            "required": true,
            "schema": {
              "min_length": 1,
              "type": "string"
            },
            "title": {
              "nullable": false,
              "required": true,
              "schema": {
                "min_length": 1,
                "type": "string"
              },
              "type": {
                "nullable": false,
                "required": true,
                "schema": {
                  "enum": ["GRAPH"],
                  "type": "string"
                }
              }
            }
          }
        }
      }
    }
  }
}
```

GraphNode

```
{
  "additional_properties": false,
  "fields": {
    "business_id": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      },
      "label": {
        "nullable": false,
        "required": true,
        "schema": {
          "enum": ["Area", "Chamber", "Equipment", "EquipmentModel", "Parameter", "ProcessStep"],
          "type": "string"
        },
        "name": {
          "nullable": false,
          "required": true,
          "schema": {
            "min_length": 1,
            "type": "string"
          },
          "node_id": {
            "nullable": false,
            "required": true,
            "schema": {
              "min_length": 1,
              "type": "string"
            },
            "properties": {
              "nullable": false,
              "required": true,
              "schema": {
                "additional_properties": true,
                "fields": {},
                "type": "object"
              }
            }
          }
        }
      }
    }
  }
}
```

GraphQueryRequest

```
{
  "additional_properties": false,
  "fields": {
    "question": {
      "nullable": false,
      "required": true,
      "schema": {
        "max_length": 1000,
        "min_length": 1,
        "type": "string"
      },
      "type": "object"
    }
  }
}
```

GraphQueryResponse

```
{
  "additional_properties": false,
  "fields": {
    "columns": {
      "nullable": false,
      "required": true,
      "schema": {
        "items": {
          "type": "string",
          "type": "array"
        },
        "error_msg": {
          "nullable": true,
          "required": false,
          "schema": {
            "type": "string"
          },
          "generated_cypher": {
            "nullable": true,
            "required": false,
            "schema": {
              "type": "string"
            },
            "is_rejected": {
              "nullable": false,
              "required": true,
              "schema": {
                "type": "boolean"
              },
              "is_valid": {
                "nullable": false,
                "required": true,
                "schema": {
                  "type": "boolean"
                },
                "latency_ms": {
                  "nullable": false,
                  "required": true,
                  "schema": {
                    "minimum": 0.0,
                    "type": "integer"
                  },
                  "question": {
                    "nullable": false,
                    "required": true,
                    "schema": {
                      "type": "string"
                    },
                    "reject_reason": {
                      "nullable": true,
                      "required": false,
                      "schema": {
                        "type": "string"
                      },
                      "row_count": {
                        "nullable": false,
                        "required": true,
                        "schema": {
                          "minimum": 0.0,
                          "type": "integer"
                        },
                      "rows": {
                        "nullable": false,
                        "required": true,
                        "schema": {
                          "items": {
                            "additional_properties": true,
                            "fields": {},
                            "type": "object"
                          },
                          "type": "array"
                        }
                      }
                    }
                  }
                }
              }
            }
          }
        }
      }
    }
  }
}
```

GraphRelationship

```
{
  "additional_properties": false,
  "fields": {
    "from_node_id": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      },
      "relation_id": {
        "nullable": false,
        "required": true,
        "schema": {
          "pattern": "^REL-[0-9a-f]{20}$",
          "type": "string"
        },
        "to_node_id": {
          "nullable": false,
          "required": true,
          "schema": {
            "min_length": 1,
            "type": "string"
          },
          "type": {
            "nullable": false,
            "required": true,
            "schema": {
              "min_length": 1,
              "type": "string"
            }
          }
        }
      }
    }
  }
}
```

GroupedMetricResult

```
{
  "additional_properties": false,
  "fields": {
    "group": {
      "nullable": false,
      "required": true,
      "schema": {
        "additional_properties": true,
        "fields": {},
        "type": "object"
      },
      "value": {
        "nullable": false,
        "required": false,
        "schema": {
          "type": "union",
          "variants": [
            {
              "type": "integer"
            },
            {
              "type": "number"
            },
            {
              "type": "null"
            }
          ]
        }
      }
    }
  }
}
```

HTTPValidationError

```
{
  "additional_properties": true,
  "fields": {
    "detail": {
      "nullable": false,
      "required": false,
      "schema": {
        "items": {
          "additional_properties": true,
          "fields": {
            "loc": {
              "nullable": false,
              "required": true,
              "schema": {
                "items": {
                  "type": "union",
                  "variants": [
                    {
                      "type": "string"
                    },
                    {
                      "type": "integer"
                    }
                  ],
                  "type": "array"
                },
                "msg": {
                  "nullable": false,
                  "required": true,
                  "schema": {
                    "type": "string"
                  },
                  "type": {
                    "nullable": false,
                    "required": true,
                    "schema": {
                      "type": "string"
                    },
                    "type": "object",
                    "type": "array"
                  }
                }
              }
            }
          }
        }
      }
    }
  }
}
```

HealthResponse

```
{
  "additional_properties": false,
  "fields": {
    "status": {
      "nullable": false,
      "required": true,
      "schema": {
        "enum": ["UP"],
        "type": "string"
      },
      "type": "object"
    }
  }
}
```

ImpactScopeBlock

```
{
  "additional_properties": false,
  "fields": {
    "check_required": {
      "nullable": false,
      "required": true,
      "schema": {
        "items": {
          "additional_properties": false,
          "fields": {
            "kind": {
              "nullable": false,
              "required": true,
              "schema": {
                "enum": ["CHAMBER", "LOT", "PARAMETER", "PROCESS_STEP", "SIBLING_CHAMBER", "WAFER"],
                "type": "string"
              },
              "relation": {
                "nullable": true,
                "required": false,
                "schema": {
                  "type": "string"
                },
                "source_id": {
                  "nullable": false,
                  "required": true,
                  "schema": {
                    "min_length": 1,
                    "type": "string"
                  },
                  "type": "object",
                  "type": "array",
                  "direct": {
                    "nullable": false,
                    "required": true,
                    "schema": {
                      "items": {
                        "additional_properties": false,
                        "fields": {
                          "kind": {
                            "nullable": false,
                            "required": true,
                            "schema": {
                              "enum": ["CHAMBER", "LOT", "PARAMETER", "PROCESS_STEP", "SIBLING_CHAMBER", "WAFER"],
                              "type": "string"
                            },
                            "relation": {
                              "nullable": true,
                              "required": false,
                              "schema": {
                                "type": "string"
                              },
                              "source_id": {
                                "nullable": false,
                                "required": true,
                                "schema": {
                                  "min_length": 1,
                                  "type": "string"
                                },
                                "type": "object",
                                "type": "array"
                              },
                              "graph_conflict": {
                                "nullable": false,
                                "required": false,
                                "schema": {
                                  "default": false,
                                  "type": "boolean"
                                },
                                "reason_code": {
                                  "nullable": true,
                                  "required": false,
                                  "schema": {
                                    "pattern": "^[A-Z0-9_]+$",
                                    "type": "string"
                                  },
                                  "status": {
                                    "nullable": false,
                                    "required": true,
                                    "schema": {
                                      "enum": ["AVAILABLE", "EMPTY"],
                                      "type": "string"
                                    }
                                  }
                                }
                              }
                            }
                          }
                        }
                      }
                    }
                  }
                }
              }
            }
          }
        }
      }
    }
  }
}
```



```
, "summary": {"nullable": true, "required": false, "schema": {"max_length": 2000, "type": "string"}}}, {"type": "object"}
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MetricPlan
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MetrologyEvidence
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NIQueryHistoryResponse
{"additional_properties": false, "fields": {"items": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"asked_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "error_msg": {"nullable": true, "required": false, "schema": {"type": "string"}}, "generated_sql": {"nullable": true, "required": false, "schema": {"type": "string"}}, "is_rejected": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "is_valid": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "latency_ms": {"nullable": true, "required": false, "schema": {"minimum": 0.0, "type": "integer"}}, "nl_query_log_id": {"nullable": false, "required": true, "schema": {"minimum": 1.0, "type": "integer"}}, "outcome": {"nullable": false, "required": true, "schema": {"enum": ["DB_ERROR", "POLICY_REJECTED", "SUCCESS", "VALIDATION_FAILED"], "type": "string"}}, "question": {"nullable": false, "required": true, "schema": {"type": "string"}}, "reject_reason": {"nullable": true, "required": false, "schema": {"type": "string"}}, "row_cnt": {"nullable": true, "required": false, "schema": {"minimum": 0.0, "type": "integer"}}, "type": "object"}, "type": "array"}, "page": {"nullable": false, "required": true, "schema": {"minimum": 1.0, "type": "integer"}}, "size": {"nullable": false, "required": true, "schema": {"maximum": 100.0, "minimum": 1.0, "type": "integer"}}, "total": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "type": "object"}
NIQueryLogItem
{"additional_properties": false, "fields": {"asked_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "error_msg": {"nullable": true, "required": false, "schema": {"type": "string"}}, "generated_sql": {"nullable": true, "required": false, "schema": {"type": "string"}}, "is_rejected": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "is_valid": {"nullable": false, "required": true, "schema": {"type": "boolean"}}, "latency_ms": {"nullable": true, "required": false, "schema": {"minimum": 0.0, "type": "integer"}}, "nl_query_log_id": {"nullable": false, "required": true, "schema": {"minimum": 1.0, "type": "integer"}}, "outcome": {"nullable": false, "required": true, "schema": {"enum": ["DB_ERROR", "POLICY_REJECTED", "SUCCESS", "VALIDATION_FAILED"], "type": "string"}}, "question": {"nullable": true, "required": true, "schema": {"type": "string"}}, "reject_reason": {"nullable": true, "required": false, "schema": {"type": "string"}}, "row_cnt": {"nullable": true, "required": false, "schema": {"minimum": 0.0, "type": "integer"}}, "type": "object"}
NIQueryOutcome
{"enum": ["DB_ERROR", "POLICY_REJECTED", "SUCCESS", "VALIDATION_FAILED"], "type": "string"}
OriginAssessment
{"additional_properties": false, "fields": {"basis": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"id": {"nullable": false, "required": true, "schema": {"max_length": 160, "min_length": 1, "type": "string"}}, "namespace": {"nullable": false, "required": true, "schema": {"enum": ["ALARM", "CHUNK", "LOT_HIST", "PARAMETER", "RELATION"], "type": "string"}}, "type": "object"}, "type": "array"}}, "compared": {"nullable": false, "required": true, "schema": {"additional_properties": false, "fields": {"downstream": {"nullable": false, "required": false, "schema": {"default": "NOT_AVAILABLE", "enum": ["CHECKED", "NOT_AVAILABLE", "NOT_CHECKED"], "type": "string"}}, "history": {"nullable": false, "required": false, "schema": {"default": "NOT_AVAILABLE", "enum": ["CHECKED", "NOT_AVAILABLE", "NOT_CHECKED"], "type": "string"}}, "metrology": {"nullable": false, "required": false, "schema": {"default": "NOT_AVAILABLE", "enum": ["CHECKED", "NOT_AVAILABLE", "NOT_CHECKED"], "type": "string"}}, "sibling": {"nullable": false, "required": false, "schema": {"default": "NOT_AVAILABLE", "enum": ["CHECKED", "NOT_AVAILABLE", "NOT_CHECKED"], "type": "string"}}, "upstream": {"nullable": false, "required": false, "schema": {"default": "NOT_AVAILABLE", "enum": ["CHECKED", "NOT_AVAILABLE", "NOT_CHECKED"], "type": "string"}}, "type": "object"}}, "degraded": {"nullable": false, "required": false, "schema": {"default": false, "type": "boolean"}}, "degraded_reasons": {"nullable": false, "required": false, "schema": {"default": [], "items": {"type": "string"}, "type": "array"}}, "dropped_basis_count": {"nullable": false, "required": false, "schema": {"default": 0, "maximum": 40, "minimum": 0, "type": "integer"}}, "scope": {"nullable": false, "required": true, "schema": {"enum": ["CURRENT_CHAMBER", "DOWNSTREAM", "EQUIPMENT_COMMON", "UNDETERMINED", "UPSTREAM"], "type": "string"}}, "type": "object"}}
```

OriginBasisRef

```
{
  "additional_properties": false,
  "fields": {
    "id": {
      "nullable": false,
      "required": true,
      "schema": {
        "max_length": 160,
        "min_length": 1,
        "type": "string"
      }
    },
    "namespace": {
      "nullable": false,
      "required": true,
      "schema": {
        "enum": [
          "ALARM",
          "CHUNK",
          "LOT_HIST",
          "PARAMETER",
          "RELATION"
        ],
        "type": "string"
      }
    }
  },
  "type": "object"
}
```

ParameterFinding

```
{
  "additional_properties": false,
  "fields": {
    "direction": {
      "nullable": false,
      "required": true,
      "schema": {
        "enum": [
          "ABOVE",
          "BELOW",
          "BOTH"
        ],
        "type": "string"
      }
    },
    "excursion_ratio": {
      "nullable": false,
      "required": true,
      "schema": {
        "type": "number"
      }
    },
    "lot_hist_ids": {
      "nullable": false,
      "required": true,
      "schema": {
        "items": {
          "type": "string"
        },
        "type": "array"
      }
    },
    "parameter_id": {
      "nullable": false,
      "required": true,
      "schema": {
        "max_length": 20,
        "min_length": 1,
        "type": "string"
      }
    },
    "step_no": {
      "nullable": false,
      "required": true,
      "schema": {
        "minimum": 1.0,
        "type": "integer"
      }
    },
    "wafer_scope": {
      "nullable": false,
      "required": true,
      "schema": {
        "enum": [
          "ALL",
          "PARTIAL",
          "SINGLE"
        ],
        "type": "string"
      }
    }
  },
  "type": "object"
}
```

ParameterItem

```
{
  "additional_properties": false,
  "fields": {
    "LCL": {
      "nullable": true,
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          "Photo"
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    },
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    },
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    }
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}
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PostActionObservationBlock

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      "required": false,
      "schema": {
        "default": "\u005c\u0085\u00815\u00801 \u00370\u00774\u00d130\u0014b\u005d0\u00b294 \u00870\u00ce58 \u00774\u00d6c4 \u00ac f5\u00815 \u00ad00\u00ce21\u00ac12\u00774 \u005c6\u005b4 \u00d6a8\u00acfc\u00b97c \u00d3c9\u00ac00\u00d560 \u00c218 \u005c6\u0074c",
        "type": "string"
      }
    },
    "status": {
      "nullable": false,
      "required": false,
      "schema": {
        "default": "NOT_AVAILABLE_STATIC_DATASET",
        "type": "string"
      }
    }
  },
  "type": "object"
}
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PublicAgentRunItem

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    "agent_run_id": {
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      "required": true,
      "schema": {
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        "type": "string"
      }
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    "alarm_id": {
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      "required": true,
      "schema": {
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        "type": "string"
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    "alarm_source": {
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      "required": true,
      "schema": {
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          "R03",
          "SUMMARY",
          "TRACE"
        ],
        "type": "string"
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    },
    "approval_id": {
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      "required": true,
      "schema": {
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        "type": "string"
      }
    },
    "chamber": {
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      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
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    },
    "confidence": {
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      "required": true,
      "schema": {
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        "minimum": 0.0,
        "type": "number"
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          "fields": {
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              "required": true,
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                  "MES"
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                "type": "string"
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              "schema": {
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                  "WAITING"
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                "type": "string"
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            }
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          "type": "object"
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        "type": "array"
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          "RFM",
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      "required": true,
      "schema": {
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      }
    },
    "fault_name": {
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      "required": true,
      "schema": {
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    },
    "status": {
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      "required": true,
      "schema": {
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          "RUNNING",
          "WAITING_APPROVAL"
        ],
        "type": "string"
      }
    },
    "tools": {
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          "fields": {
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              "required": true,
              "schema": {
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              "required": true,
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                "type": "string"
              }
            },
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              "schema": {
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                "type": "string"
              }
            }
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          "type": "object"
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        "type": "array"
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    }
  },
  "type": "object"
}
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PublicApprovalDecision

```
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    "REJECTED"
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  "type": "string"
}
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PublicApprovalItem

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  "fields": {
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      "required": true,
      "schema": {
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          "MONITORING",
          "WARNING"
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        "type": "string"
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    },
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      "required": true,
      "schema": {
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        "type": "string"
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      "nullable": false,
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      "schema": {
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    "approval_id": {
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      "required": true,
      "schema": {
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        "type": "string"
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    },
    "approved_at": {
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    "approved_by": {
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      "schema": {
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    },
    "chamber": {
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      "required": true,
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}
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